

Xytron™ M6516A

PPS-(GF+MD)65

Injection Molding, 65% Glass/Mineral Reinforced, Excellent Dimensional Stability

Print Date: 2021-12-24

Properties	Typical Data	Unit	Test Method
Rheological properties			
	Value		
Molding shrinkage (parallel)	0.3	%	ISO 294-4
Molding shrinkage (normal)	0.55	%	ISO 294-4
Mechanical properties			
	Value		
Tensile modulus	21500	MPa	ISO 527-1/-2
Tensile modulus (80°C)	18500	MPa	ISO 527-1/-2
Tensile modulus (160°C)	10500	MPa	ISO 527-1/-2
Tensile modulus (180°C)	9650	MPa	ISO 527-1/-2
Tensile modulus (200°C)	9550	MPa	ISO 527-1/-2
Stress at break	150	MPa	ISO 527-1/-2
Stress at break (80°C)	135	MPa	ISO 527-1/-2
Stress at break (160°C)	83	MPa	ISO 527-1/-2
Stress at break (180°C)	72	MPa	ISO 527-1/-2
Stress at break (200°C)	65	MPa	ISO 527-1/-2
Strain at break	1	%	ISO 527-1/-2
Strain at break (80°C)	1	%	ISO 527-1/-2
Strain at break (160°C)	2.4	%	ISO 527-1/-2
Strain at break (180°C)	2.3	%	ISO 527-1/-2
Strain at break (200°C)	2.1	%	ISO 527-1/-2
Flexural modulus	21000	MPa	ISO 178
Flexural modulus (80°C)	19500	MPa	ISO 178
Flexural modulus (120°C)	12500	MPa	ISO 178
Flexural modulus (160°C)	10000	MPa	ISO 178

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Property Data (Provisional)

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Properties	Typical Data	Unit	Test Method
Flexural modulus (200°C)	8800	MPa	ISO 178
Flexural strength	260	MPa	ISO 178
Flexural strength (80°C)	225	MPa	ISO 178
Flexural strength (120°C)	160	MPa	ISO 178
Flexural strength (160°C)	120	MPa	ISO 178
Flexural strength (200°C)	90	MPa	ISO 178
Charpy impact strength (+23°C)	25	kJ/m ²	ISO 179/1eU
Charpy impact strength (-30°C)	25	kJ/m ²	ISO 179/1eU
Charpy notched impact strength (+23°C)	8	kJ/m ²	ISO 179/1eA
Charpy notched impact strength (-30°C)	8.5	kJ/m ²	ISO 179/1eA
Izod impact strength (+23°C)	24	kJ/m ²	ISO 180/1U
Izod impact strength (-40°C)	23	kJ/m ²	ISO 180/1U
Izod notched impact strength (+23°C)	8.1	kJ/m ²	ISO 180/1A
Izod notched impact strength (-40°C)	8.8	kJ/m ²	ISO 180/1A
Weldline strength at thickness 1	42	MPa	ISO 527-1/-2
Weldline strain at thickness 1	0.3	%	ISO 527-1/-2
Thickness tested (1)	4	mm	ISO 527-1/-2

Thermal properties

Value

Melting temperature (10°C/min)	280	°C	ISO 11357-1/-3
Glass transition temperature (10°C/min)	90	°C	ISO 11357-1/-2
Temp. of deflection under load (1.80 MPa)	275	°C	ISO 75-1/-2
Coeff. of linear therm. expansion (parallel)	0.15	E-4/°C	ISO 11359-1/-2
Coeff. of linear therm. expansion (normal)	0.3	E-4/°C	ISO 11359-1/-2
Coef. of lin. therm expansion, parallel, above Tg	0.12	E-4/°C	ISO 11359-1/-2
Coef. of lin. therm expansion, normal, above Tg	0.8	E-4/°C	ISO 11359-1/-2

Electrical properties

Value

Volume resistivity	>1E13	Ohm*m	IEC 62631-3-1
Surface resistivity	>1E15	Ohm	IEC 62631-3-2

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Property Data (Provisional)

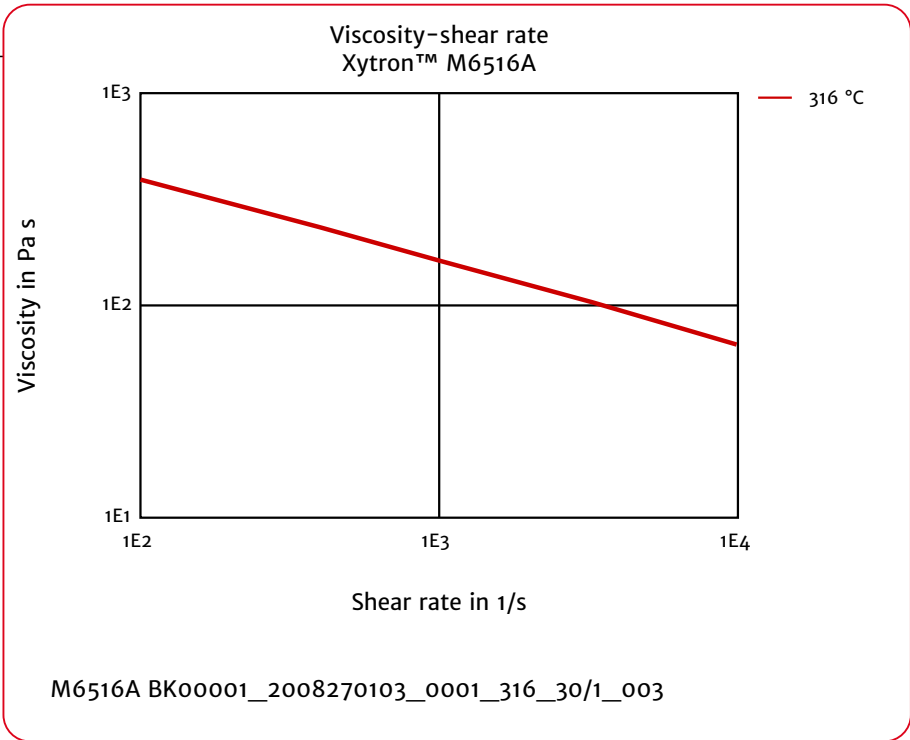
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Properties	Typical Data	Unit	Test Method
Other properties	Value		
Density	1970	kg/m³	ISO 1183

Viscosity-shear rate



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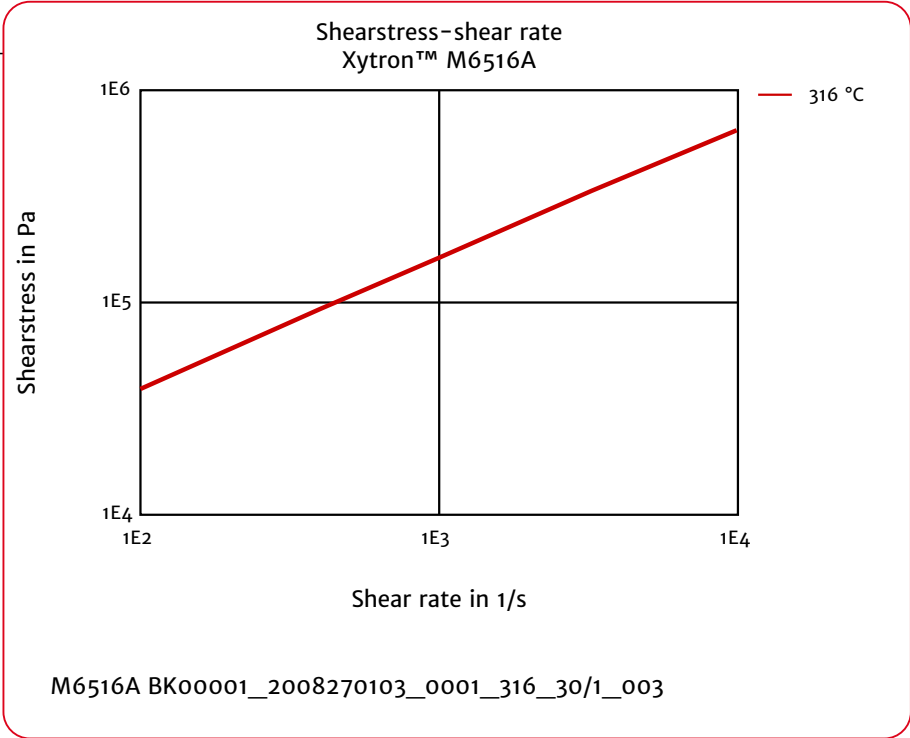
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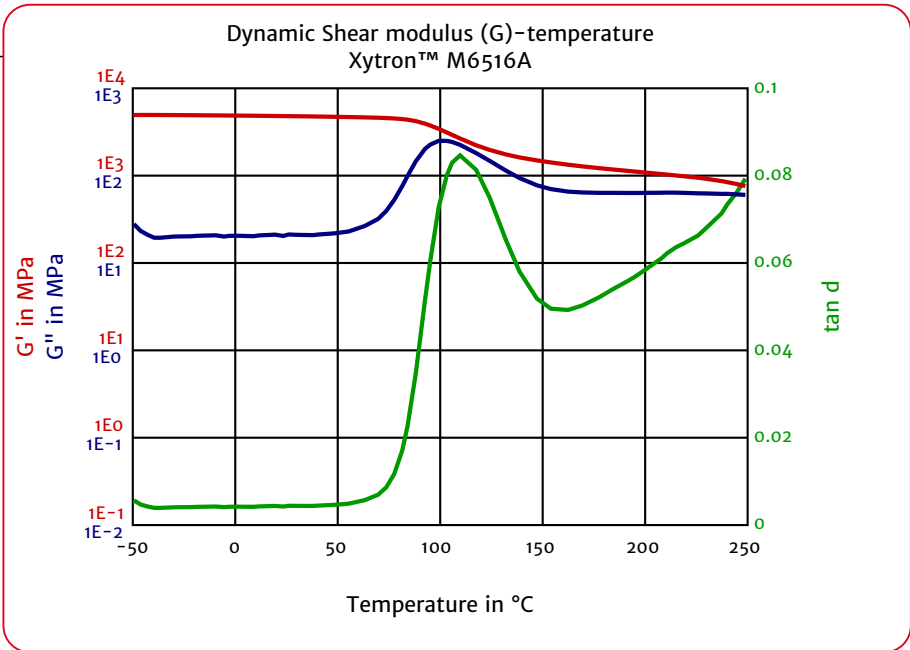
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Shearstress-shear rate



Dynamic Shear modulus
(G)-temperature



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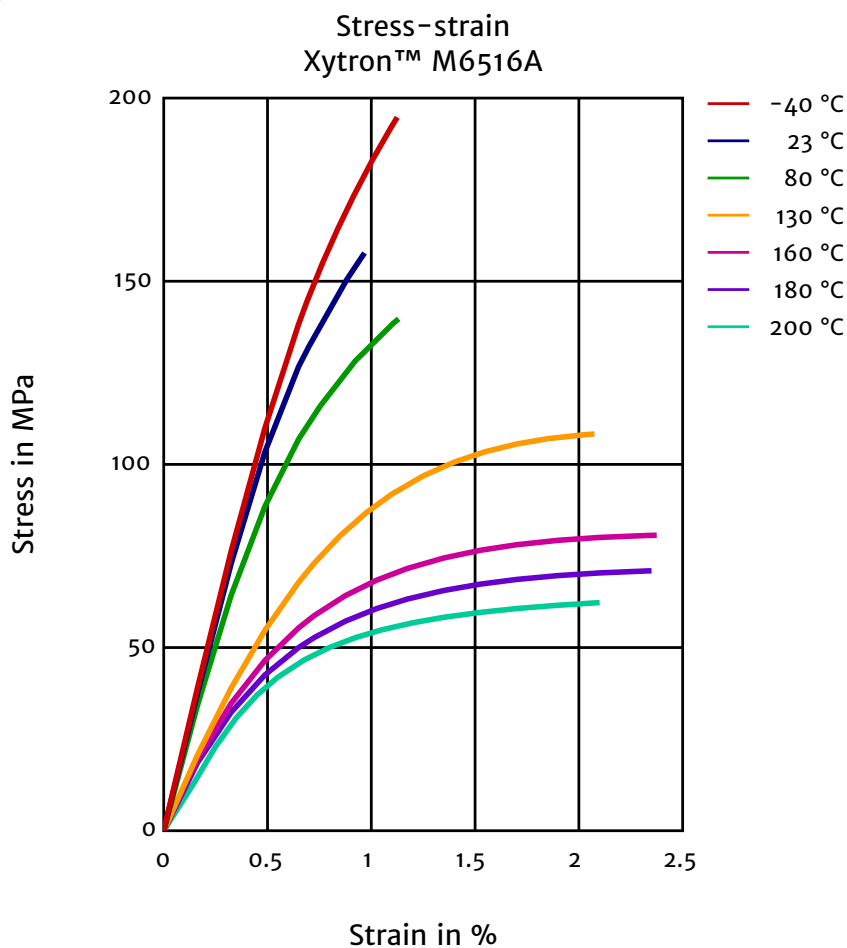
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Stress-strain



Job_2020_02965

Job_2020_04093

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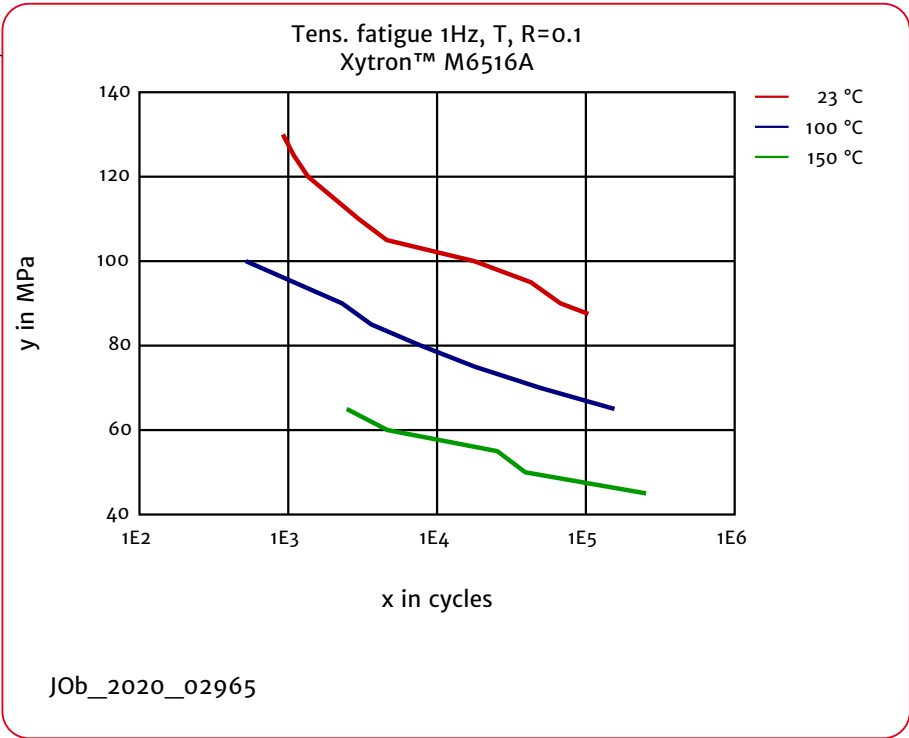
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Tens. fatigue 1Hz, T, R=0.1



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